Problem Set 2

Given on: 29th October 2024
Due on: 15th November 2024
[No late submissions shall be accepted]
Total: **50 points**
Submission Via: Hard Copy, no soft copies will be checked
Topics: Pumping Lemma, CFGs & Turing Decidable Languages

Question 1 [20 points]

Give 2 examples of languages that are:

(a) regular

(b) context-free but not regular

(c) Turing-decidable but not context-free

(d) Turing-recognizable but not decidable

(e) not Turing-recognizable

Question 2 [10 points]

Prove that the class of context-free languages is not closed under either intersection or complement. [Hint: the class of context-free languages is known to be closed under union.]

Question 3 [10 points]

Design a Turing Machine to decide the language $L = \{ww^{rev} : w \epsilon \{a, b\}\}$, here w^{rev} represents the reverse of the string w.

Question 4 [10 points]

Let L be a language over some Σ^* defined as $L = \{w : w = w^{rev}\}$. Show that L is not regular using the pumping lemma. [L is the language of palindromes.]